Circuit Lab B/C Test

January 10, 2026

Greenbrier Invitational

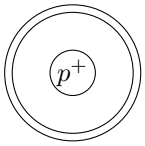
Written By: Shiven Gupta, Singapore Armed Forces '27

Page:	2	3	4	Total
Points:	18	18	24	60
Score:				

1. Do **NOT** write on this packet. Only answer on your answer sheet.
2. The points per question are boxed in the margin next to the question.
3. You may separate the test, but ensure you put it back in order.
4. You may remove materials from your binder ***AFTER*** the time has started.
5. **SHOW ALL OF YOUR WORK**
6. Tiebreaker questions have TB#(order) in front of them. (QN6, QN18, QN22)

Multiple Choice [36 pts]

1. [2] A point charge is placed in a two-walled hollow sphere. What is the net surface charge of the hollow sphere?



   A. p^+
   B. $2p^+$
   C. $\dfrac{p^+}{4\pi r^2}$
   D. 0

2. [2] When should you ground yourself?
   A. When you behave really badly.
   B. When working with sensitive electronics.
   C. When working with electrical sockets.
   D. When you need to reduce stress and panic.

3. [2] Ohm is a derived unit. Which base units is it made of?
   A. $kg,\ s,\ m,\ V$
   B. $A,\ s,\ V,\ m$
   C. $kg,\ V,\ m,\ A$
   D. $kg,\ s,\ m,\ A$

4. [2] Identify the below symbol.



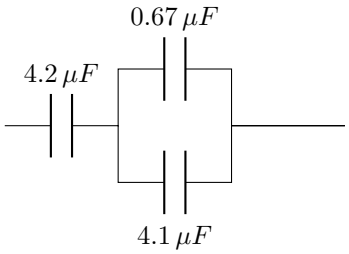
   A. Variable Resistor
   B. Potentiometer
   C. Resistor
   D. None of the Above

5. [2] How does an electromagnetic wave propagate?
   A. Perpendicular to both the electric field and magnetic field
   B. Perpendicular to the electric field and parallel to the magnetic field
   C. Parallel to the electric field and perpendicular to the magnetic field
   D. Parallel to both the electric field and magnetic field

6. [2] TB#1 What is the magnetic flux through a conductor with potential difference $\epsilon\ (V)$, length $\ell\ (m)$, resistivity $\rho\ (\Omega m)$, radius $r\ (m)$? Assume that it is also connected to a capacitor in the circuit, and that the effective capacitance is $C\ (F)$ and the resonant frequency is $\omega\ (rad/s)$.
   A. $\Phi_B = \dfrac{\epsilon\pi r^2}{\rho\ell\omega C}$ (Wb)
   B. $\Phi_B = \dfrac{\epsilon}{\rho\ell\omega^2 C}$ (Wb)
   C. $\Phi_B = \dfrac{\epsilon\pi r^2}{\rho\ell\omega^2 C}$ (Wb)
   D. $\Phi_B = \dfrac{\epsilon\pi r^2\omega^2 C}{\rho\ell}$ (Wb)

7. [2] What is the ferromagnetic ability of a metal reliant on?
   A. Lattice structure of metals
   B. Alignment of atomic magnetic movements
   C. Polarity Differences and Electronegativity
   D. Vacuum permeability (μ_0)

8. [2] Resolve the equivalent capacitance of the junction.



   A. $4.78\,\mu F$
   B. $2.23\,\mu F$
   C. $0.506\,\mu F$
   D. $8.97\,\mu F$

9. [2] Who formulated the law that describes the force between two point charges?
   A. André-Marie Ampère
   B. Charles-Augustin de Coulomb
   C. Michael Faraday
   D. Nikola Tesla

10. [2] Which of the following **cannot** be considered as real-life Faraday cages?
    A. None of the Listed
    B. Car Chassis
    C. Microwave Ovens
    D. Elevator

11. [2] Which scientist is credited with inventing the electric battery and demonstrating that electricity could be produced chemically?
    A. Georg Ohm
    B. Gustav Kirchhoff
    C. Alessandro Volta
    D. James Clerk Maxwell

12. [2] Maxwell-Faraday's Law can be written as (in differential form):
    $\vec{\nabla}_x \times \vec{E} = \frac{\partial \vec{B}}{\partial t}$.
    What does it state?
    A. A time-varying magnetic field always accompanies a spatially varying non-conservative electric field.
    B. The coil of a spring is proportional to the electric field it generates.
    C. The change in electric field is proportional to the time-varying magnetic field.
    D. The change in electric field is proportional to the time-varying magnetic force.

13. [2] Which of the following is theoretically not a possible switch?
    A. Single-Pole Single-Throw (SPST)
    B. Double-Pole Double-Throw (DPDT)
    C. Triple-Pole Quadruple-Throw (TPQT)
    D. None of the Above

14. [2] Which device is designed to automatically disconnect a circuit when excessive current flows, preventing damage to the wiring?.
    A. Switch
    B. Relay
    C. Circuit breaker
    D. Ground fault circuit interrupter

15. [2] Which of the following devices protects people from electric shock by detecting leakage currents as small as a few milliamperes and disconnecting the circuit almost instantly?
    A. Fuse
    B. Circuit breaker
    C. Ground fault circuit interrupter (GFCI)
    D. Electromagnetic relay

16. [2] What is a common source of direct current?
    A. AC generator
    B. Transformer
    C. Battery
    D. Alternator

17. [2] Which of the following is true of alternating current?
    A. 3-phase is stronger than 1-phase.
    B. It changes direction periodically (specifically following a tangential time scale)
    C. It is used for high-voltage transmission over long distances
    D. It has heavy voltage line losses.

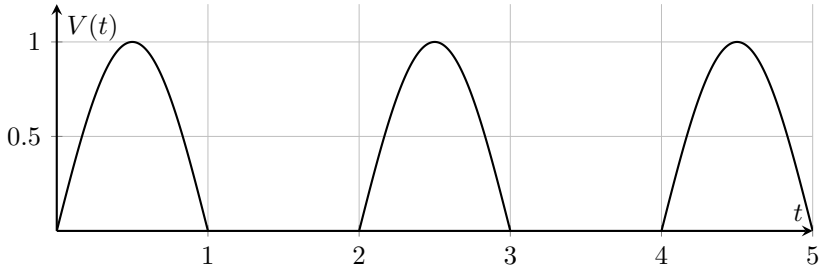
18. [2] TB#2 What is the resistance of the below resistor?
    (In case of printing errors: blue, red, orange, brown, green, gray)



    A. $632\,\Omega \pm 0.5\%$
    B. $6230\,\Omega \pm 0.5\%,\ 1\,ppm/K$
    C. $623\,\Omega \pm 1\%,\ 2\,ppm/K$
    D. $62300\,\Omega \pm 0.5\%,\ 1\,ppm/K$

Free Response [24 pts]

19. [2] The following graph depicts the AC voltage output passed through a half-wave rectifier. Draw the effect on the waveforms if the voltage output is now passed through a full-wave rectifier.



20. [2] What is electron current, and at what speed does it occur?

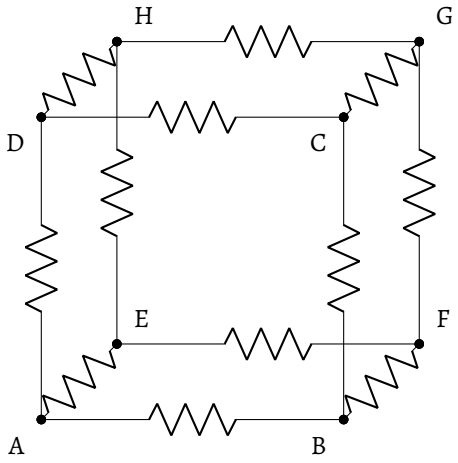
21. [5] Define energy flux in a circuit. How does it travel? Where does it do work? What is it measured in? What is it described by?

22. [3] TB#3 Which forms of energy is electrical energy dissipated through when passing through buzzers?

23. [2] The effective resistance of a circuit is $28.3\,\Omega$. What is the real component of the impedance $Z_{\text{re}} \in \mathbb{R}$?

24. [4] Approximately how many time constants τ_{RC} are needed for a capacitor in an RC circuit to reach 47% of its full charge?

25. [6] There exists a resistor cube. Solve for the resistance between A and G. Assume all resistors are of value 1Ω.



GREENBRIER CIRCUIT B/C LAB || HANDS-ON ACTIVITY

This portion will be 25% of your overall score.

TEAM NAME & #: _____

You are given

a) ∞ Ideal Wire	d) 555 Timer IC	g) 3x 330Ω-1kΩ Resistors	j) 3x diodes
b) 5V Battery	e) Decade Counter	h) 2x 10kΩ-100kΩ Resistors	k) 3x NPN transistors
c) Switch	f) 3x LEDs (G, Y, R)	i) 2x 1-100μF capacitors	l) Breadboard

A traffic light goes from Green → Yellow → Red and repeats with Green. Using the above materials, design a traffic light system that alternates between the colors appropriately. Linger on each color for 2-3 seconds. Label all components appropriately.
Feel free to use the next page for scratch work.

Team's Final Answer:

SCRATCH WORK: